

Performance Metrics used in Linear Regression.

i. R Squared

ii. Adjusted R Squared

i. R squared

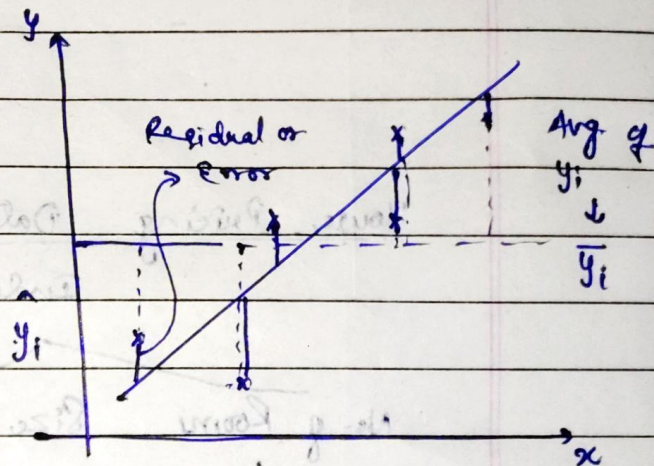
$$R^2 = 1 - \frac{SS_{res}}{SS_{TOTAL}}$$

$$= 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y}_i)^2}$$

$$= 1 - \frac{\text{Small number}}{\text{Bigger number}}$$

$$= 1 - (\text{Small number})$$

$$\approx 1$$



1) Accurate my model is?

0.70 $\Rightarrow$ 70%

0.80 $\Rightarrow$ 80%

0.90 $\Rightarrow$ 90%

{ overfitting, underfitting }

ii. Adjusted R Squared.

Dataset $\rightarrow$ (Price)

Size of the house $\uparrow$ Price $\uparrow$

(Gender)

Size of the house

No. of bedrooms

Location

Price.

the correlation

R Squared $\approx$ 75% $=$ 0.75

R Squared $\approx$ 80% $=$ 0.80

No. of bedrooms $\uparrow$ Price $\uparrow$

the correlation

R^2 squared $\Rightarrow 85\% = 0.85$

R^2 squared $\Rightarrow 87\% = 0.87$

Note: This is the problem of R^2 squared even though you don't add an input feature which is directly correlated with output feature then also the value will increase.

To resolve this we use Adjusted R^2 squared.

$$\text{Adjusted } R^2 \text{ squared} = 1 - \frac{(1 - R^2)(N - 1)}{N - P - 1}$$

N = No. of data points

P = No. of Independent feature

$P = 2$ $R^2 = 90\%$ $R^2 \text{ adjusted} = 86\%$

$P = 3$ $R^2 = 92\%$ $R^2 \text{ adjusted} = 82\%$

Note: If the additional independent feature is directly correlated with output feature the Adjusted R^2 squared value will increase with previous ones, if not then decrease with previous ones.

(In the case of addition independent feature is not directly correlated with output feature).